

Advances in Deep Learning Architectures: A Survey

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Abstract

This paper presents a survey of recent advances in the field. We review key contributions, discuss methodological trends, and identify open challenges. Our analysis covers both foundational work and recent developments, highlighting directions for future research.

1. Introduction

Deep learning has transformed the field of artificial intelligence over the past decade. Convolutional neural networks were first applied to large-scale image classification by Krizhevsky et al. [5], achieving unprecedented accuracy on the ImageNet benchmark. The introduction of residual connections by He et al. [2] allowed training of networks with hundreds of layers, overcoming the vanishing gradient problem. More recently, attention mechanisms [1] have replaced recurrence as the dominant sequence modelling paradigm.

2. Related Work

Generative modelling has progressed rapidly since the introduction of adversarial training [3]. Optimization advances such as adaptive gradient methods [4] have become standard practice. Graph-based attention approaches have been extended to multi-modal settings [6], while sparse attention variants have been explored in recurrent frameworks [7]. Neural architecture search methods based on probabilistic models [8] aim to automate model design. Self-supervised pretraining for dense prediction tasks [9] reduces the need for labelled data, and token reduction strategies [10] improve inference efficiency of transformer models.

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